In the supposition below:
1) Is the interaction between protons and electrons mediated by magnetrons, not by virtual photons, as QED Feynman diagrams claim?-ok
2) Feynman diagrams must be calculated without adding all theoretically possible interactions because interactions that have another particle carrying force other than virtual photons must be canceled, and those that are not possible (which do not resonate according to Quark Information Theory) must have a coefficient Ci=0 for any i=1...n in Quantum Electrodynamics power series $g = 1+C1(\alpha/\pi)+C2(\alpha/\pi)^{2+C3}(\alpha/\pi)^{3+C4}(\alpha/\pi)^4 +C5(\alpha/\pi)^5$ ...

+... I don't know if it converges to a fixed value, but it is limited to a closed range. !?

3) Are Feynman diagrams correct in the case of magnetic monopole interactions where the above condition is not met? And the answer is probably yes, because it provides a correct value. It is also shown that the monopole provides the same value in that case, but from now on we must develop that direction of the monopole. Ok

Three Methods for the p–e⁻ Electromagnetic Interaction: QED, Dual QED (Zwanziger), and the TauTimp Monopole — with Feynman Diagrams, TauTimp / Quark Information Theory framework September 11, 2026

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1 Three Methods for the p–e⁻ Electromagnetic Interaction: QED, Dual QED (Zwanziger), and the TauTimp Monopole

1.1 0. What this document is

This document **unifies** the two earlier notes in this directory into a single, complete academic treatment, with **Feynman diagrams for all three calculation methods** of the p–e⁻ interaction:

- **Method 1** — the virtual photon (standard QED), with the already-established diagrams (Fig. 1–3).
- **Method 2** — dual QED / the Zwanziger formalism (two potentials A_μ, B_μ), treated so far only textually/tabularly in *Feynman_QED_duala_Zwanziger_EN.md*; this document adds the **missing Feynman diagrams** (Fig. 10–11).
- **Method 3** — the magnetic monopole/"magnetron" TauTimp hypothesis, corrected variant (resonance + K-matrix unitarization), with the already-established diagrams (Fig. 4, 5, 7, 8, 9).

Companion documents in this directory:

- *Feynman_photon_and_corrected_monopole_EN.md* / *Feynman_foton_si_monopol_corectat_RO.md* — the two-method consolidated document (1 and 3); this document **does not replace it**, it carries over its §1–§2 diagrams and results almost unchanged and places them alongside Method 2.
- *Feynman_QED_duala_Zwanziger_EN.md* / *_RO.md* — the document that first introduced the three-model comparison (mediator/source table), without diagrams; this document is its natural continuation, adding the graphical representation.
- *Feynman_diagrams_photon_monopole_EN.md*, *Feynman_monopole_unitarization_EN.md* — the step-by-step derivations of Method 3.
- *monopole_wave_equation_full.md* — the Bessel spectrum of the TauTimp field, the basis of Method 3's resonance rule.
- *s10052-018-6440-6.pdf* — S. Baines, N.E. Mavromatos, V.A. Mitsou, J.L. Pinfold, A. Santra, *Monopole production via photon fusion and Drell–Yan processes…*, Eur. Phys. J. C (2018) 78:966 — the source of the dualization formalism used in Method 2 and of the $g(\beta), \kappa$ perturbativity mechanism cited in §7.
- *Feynman.docx* — the original questions motivating the whole line of analysis.
- *figures/* — all PNG figures used below; Fig. 10, 11 and 12 are new, generated for this document (see the companion skill's Appendix for the generation script).
- *Feynman_ploturi_structurate_trei_metode.mpl* — companion Maple transcript, with a structured 3×3 dashboard (vertex/process diagram / observable panel, for each of the three methods) visualizing the diagrams below directly in Maple; extends *Feynman_ploturi_structurate_foton_monopol.mpl* (which only covered Methods 1 and 3) with the Method 2 row.

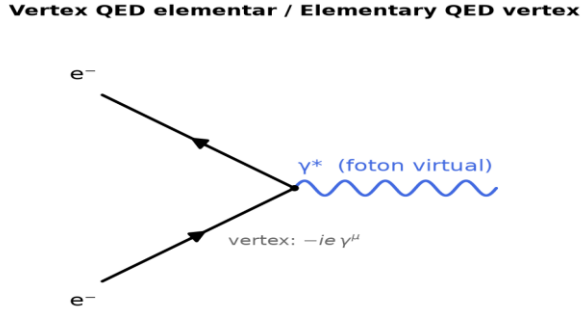
Scientific-honesty note, valid throughout, on three distinct levels: **Method 1** is standard QED, experimentally confirmed to 10^{-12} precision. **Method 2** is an established, published field-theory formalism (Zwanziger 1971), actively used in real experimental monopole searches at the LHC — but for an unobserved particle. **Method 3** is a speculative framework of this project (TauTimp), formally coherent with the first two by construction, but without the status of a tested theory. Do not conflate these levels.

1.2 1. Method 1 — The virtual photon (standard QED) *1.2.1* 1.1 Feynman rules

$$\mathcal{L}_{int} = -e\,\bar{\psi}\gamma^\mu\psi\,A_\mu$$

Element	Rule
Fermion propagator	$\frac{i(\gamma\cdot p + m)}{p^2 - m^2 + i\varepsilon}$
Photon propagator (Feynman gauge)	$\frac{-ig_{\mu\nu}}{q^2 + i\varepsilon}$
Electron–photon vertex	$-ie\,\gamma^\mu$

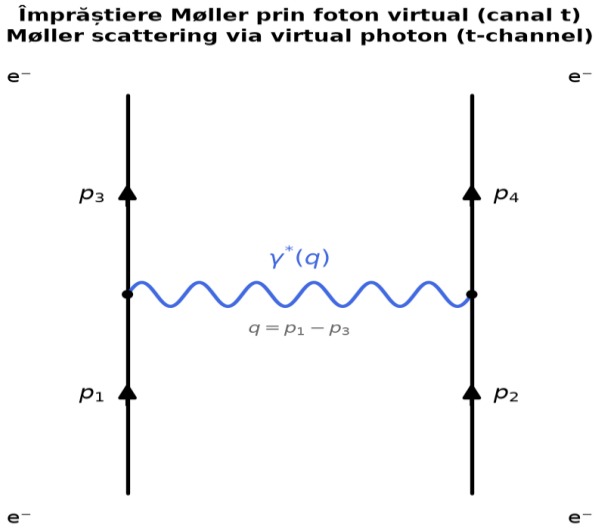
Fig. 1 — Elementary QED vertex.



QED vertex

1.2.2 1.2 Møller diagram ($e^-e^- \to e^-e^-$, t-channel)

Fig. 2 — Virtual photon exchange.

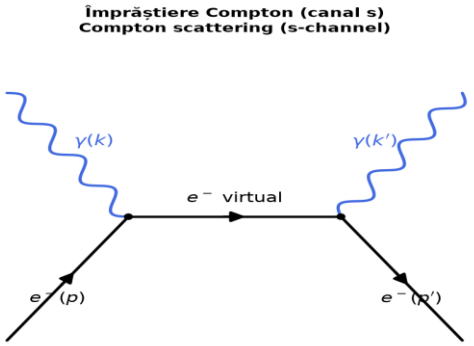


Moller photon

$$i\mathcal{M}_t = \bar{u}(p_3)(-ie\gamma^\mu)u(p_1)\ \frac{-ig_{\mu\nu}}{q^2}\ \bar{u}(p_4)(-ie\gamma^\nu)u(p_2), \qquad q = p_1 - p_3$$

1.2.3 1.3 Compton diagram

Fig. 3 — Virtual electron propagator, *s*-channel.



Compton

1.2.4 1.4 Coupling and the perturbative series

$$\alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c} \approx \frac{1}{137.036}$$

The series $g = 1 + C_1(\alpha/\pi) + C_2(\alpha/\pi)^2 + \cdots$ is only **asymptotic** (Dyson 1949), not strictly convergent — but $\alpha/\pi \approx 2.3 \times 10^{-3}$ is small enough that the Born level (a single photon exchange) is already an excellent approximation.

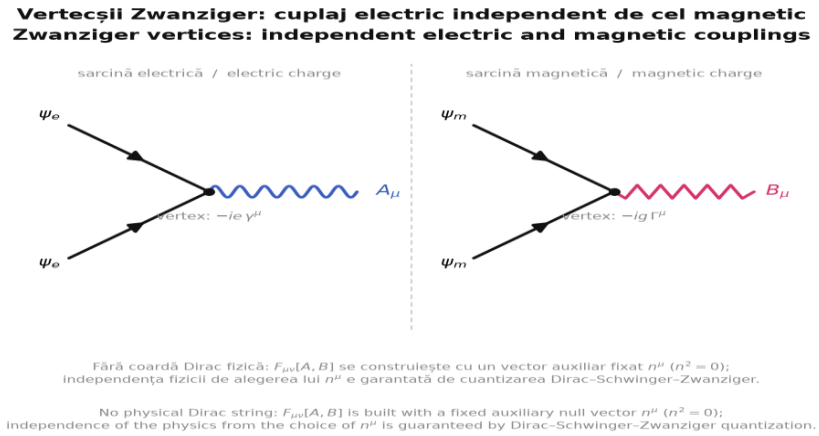
1.3 2. Method 2 — Dual QED / the Zwanziger formalism *1.3.1* 2.1 Why two potentials are needed

A static magnetic monopole produces $\vec{B} \propto g\hat{r}/r^2$, which cannot arise from a single regular potential A_μ everywhere in space ($\nabla \cdot \vec{B} \neq 0$ would contradict $\vec{B} = \nabla \times \vec{A}$). Dirac's solution (1931) — a single A_μ, singular along a string — requires the quantization condition

$$e\,g = \frac{n}{2}\,(4\pi\epsilon_0 c)\,\xi\,\hbar c, \qquad n \in \mathbb{Z}$$

(cf. eq. (1), s10052-018-6440-6, p. 2), which in natural SI units gives $g = n\,e/(2\alpha) \equiv n\,g_D$, $g_D \approx 68.5\,e$.
Zwanziger (1971) removes the string from a local-Lagrangian formulation with **two** potentials: electric charges couple to A_μ, magnetic charges to B_μ, and the field-strength tensors are built using a fixed, lightlike auxiliary vector n^μ ($n^2 = 0$) — independence of the physics from the choice of n^μ is guaranteed by Dirac–Schwinger–Zwanziger quantization.

Fig. 10 — Zwanziger vertices: independent electric and magnetic couplings.

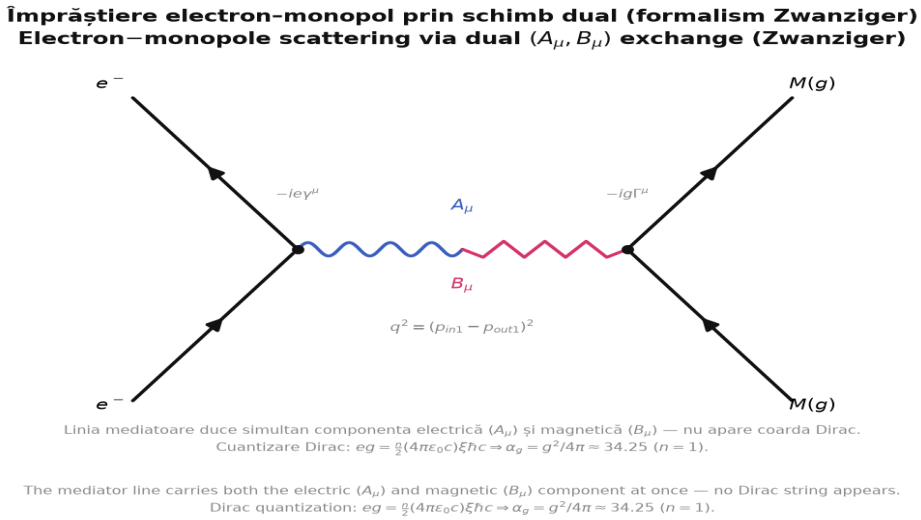


Zwanziger vertices

Unlike the TauTimp monopole vertex (Fig. 4, §3.1 below), **no physical Dirac string appears here** — the two vertices are independent, each coupled to its own potential.

1.3.2 2.2 The electron–monopole scattering diagram

Fig. 11 — e^-–monopole scattering via dual (A_μ, B_μ) exchange.



Electron-monopole Zwanziger

The mediator line carries both the electric and magnetic component at once — a single channel, formed from the two A_μ/B_μ halves of the dual propagator, with no visible string in the diagram.

1.3.3 2.3 What the monopole-search literature actually uses

Our primary source, s10052-018-6440-6, does **not** dualize the full Lagrangian with B_μ; it uses the simpler, single-potential substitution:

$$e \;\mapsto\; g, \qquad \alpha_e = \frac{e^2}{4\pi} \;\mapsto\; \alpha_g = \frac{g^2}{4\pi}$$

(cf. §1, s10052-018-6440-6). Fig. 10–11 above illustrate the Zwanziger formalism conceptually (mediator/source, absence of the string), not a line-by-line re-derivation of the full Lagrangian — which is not reproduced from any source available locally.

1.3.4 2.4 Quantization remains non-perturbative

$g_D \approx 68.5\,e \Rightarrow \alpha_g = g^2/4\pi \approx 34.25$ $(n=1)$ — **non-perturbative**, unlike $\alpha \approx 1/137$. The central reason neither Method 2 nor Method 3 can be treated as a truncated Born series.

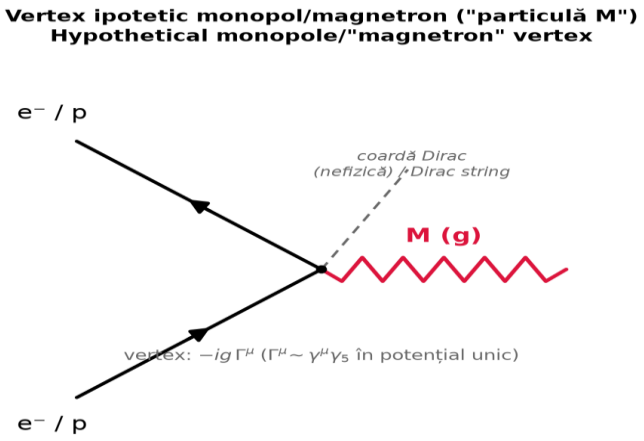
1.4 3. Method 3 — The TauTimp monopole ("the magnetron"), corrected variant

1.4.1 3.1 Why the bare Dirac coupling cannot be used directly

$$eg = 2\pi n\hbar c \quad\Rightarrow\quad \alpha_g = \frac{g^2}{4\pi} = \frac{1}{4\alpha} \approx 34.25 \;\; (n=1)$$

is a **topological necessity** (invisibility of the Dirac string), not an adjustable parameter.

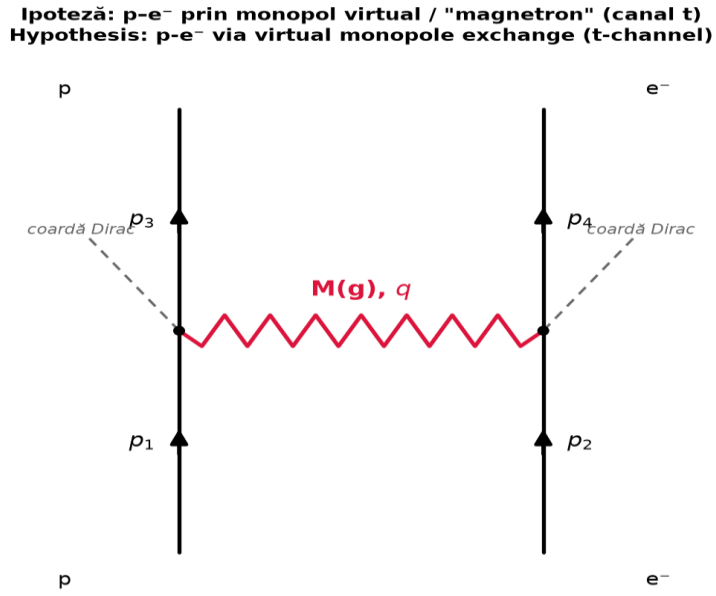
Fig. 4 — Elementary monopole vertex, with the Dirac string.



Monopole vertex

1.4.2 3.2 The p–e⁻ monopole-hypothesis diagram

Fig. 5 — Hypothesis: p–e⁻ via virtual monopole exchange (t-channel).



p-e monopole

1.4.3 3.3 The resonance rule

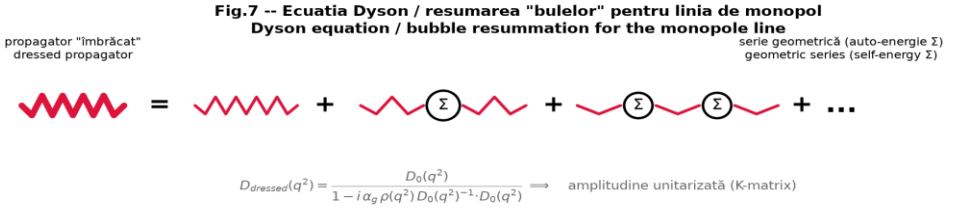
From *monopole_wave_equation_full.md* (§6), the radial modes of the TauTimp field are quantized on the zeros of J_0:

$$k_{r,n} = \frac{j_{0,n}}{2R}, \qquad j_{0,1} \approx 2.4048,\ j_{0,2} \approx 5.5201, \ldots \qquad R = 4.2355 \times 10^{22}\ \text{m}$$

Rule: the channel is allowed ($C_i \neq 0$) **only if** $q^2 = k_{r,n}^2$; otherwise $C_i = 0$ — exactly the rule required in *Feynman.docx* (questions 2–3).

1.4.4 3.4 Dyson resummation / K-matrix unitarization

Fig. 7 — The Dyson equation for the monopole line.



Dyson equation

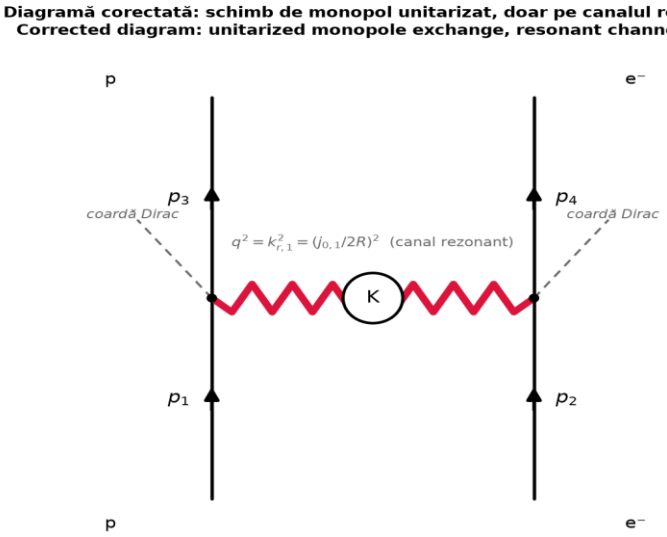
$$D_{dressed}(q^2) = \frac{D_0(q^2)}{1 - [-i\Sigma(q^2)]D_0(q^2)} \quad \Rightarrow \quad a(q^2) = \frac{K(q^2)}{1 - iK(q^2)}, \quad K = \alpha_g\, \rho(q^2)$$

$$\boxed{|a(q^2)|^2 = \frac{K^2}{1+K^2}}$$

Theorem: For any finite real K, $|a|^2 < 1$ strictly. The physical coupling is **subunitary by construction**.

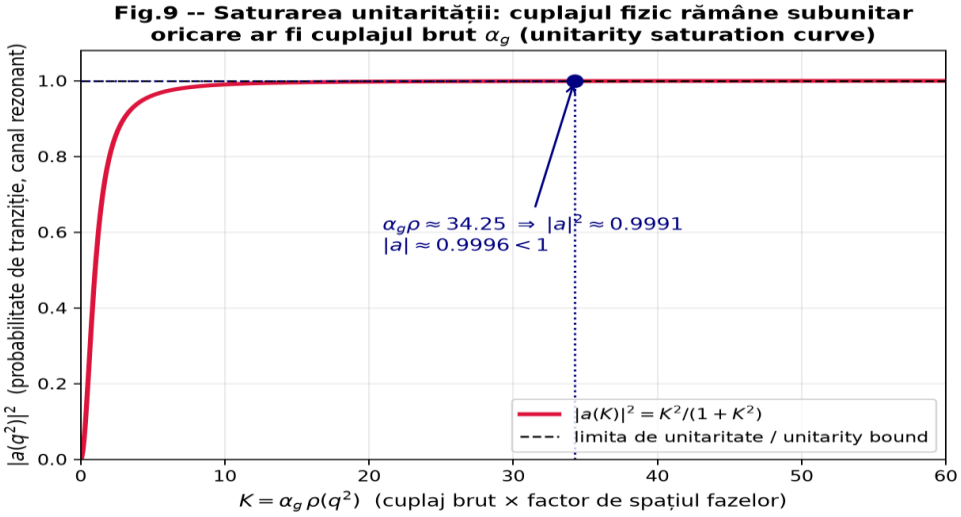
1.4.5 3.5 The final corrected diagram and numerical check

Fig. 8 — Corrected diagram: unitarized monopole exchange, resonant channel.



Corrected diagram

Fig. 9 — Unitarity saturation curve.

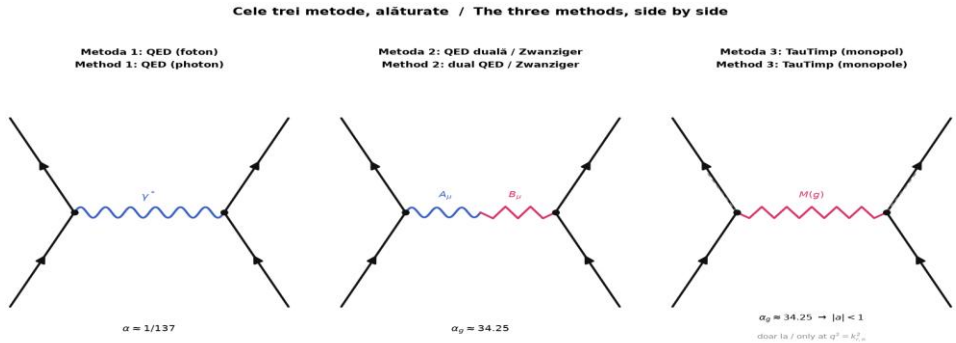


Unitarity curve

With $\alpha_g \approx 34.25$ $(n = 1)$ and $\rho \approx 1$ (on resonance): $K \approx 34.25 \Rightarrow |a|^2 \approx 0.99915 \Rightarrow |a| \approx 0.9996 < 1$.

1.5 4. Visual comparison — the three methods side by side

Fig. 12 — Method 1 (QED), Method 2 (Zwanziger), Method 3 (TauTimp), identical external legs, different mediators.



Three methods

All three diagrams share the same external fermion legs; what differs is only the mediator line — continuous blue (Method 1), half blue/half red with no string (Method 2), red zigzag with a Dirac string and restricted to resonances (Method 3).

1.6 5. Extended comparison table

	Method 1 — QED	Method 2 — Dual QED/Zwanziger	Method 3 — TauTimp (monopole)
Mediator	photon γ (A_μ)	A_μ and B_μ (or dualized A_μ)	monopole M (resonant TauTimp mode)
Source	electric charge e	electric e + magnetic g charges, independent	electric charge/field e — same as Method 1
Vertex diagram	Fig. 1	Fig. 10	Fig. 4
Process diagram	Fig. 2, 3	Fig. 11	Fig. 5, 8
Bare coupling	$\alpha = e^2/4\pi \approx 1/137$	$\alpha_g = g^2/4\pi \approx 34.25$	$\alpha_g \approx 34.25$ (identical, via Dirac quantization)
Domain of q^2	continuous	continuous (formally)	discrete: $q^2 = k_{r,n}^2$
Perturbativity mechanism	inherent ($\alpha \ll 1$)	$g(\beta) = g\beta^\delta$, $\kappa = 1$ (spin-1 unitarity)	resonance rule + K-matrix unitarization
Dirac string	—	absent from the construction	present (Fig. 4), unphysical
Experimental status	confirmed (10^{-12})	used in LHC/MoEDAL searches; unobserved	in-house speculative framework, unobserved

1.7 6. Check against *Feynman.docx*

Rule in *Feynman.docx*	How it is satisfied
(1) p–e⁻ interaction via magnetrons, not only photons	Method 3, §3.2: a distinct monopole-exchange channel, sourced by the same electric charge
(2) $C_i = 0$ for non-resonant channels	the Bessel resonance rule, §3.3
(3) diagrams are incorrect without the above condition	The corrected diagram (Fig. 8) imposes exactly that condition
(5) the QED (Dyson) series does not converge to a fixed value	§1.4, the motivation for exact resummation

1.8 7. Discussion

The three methods represent **three distinct levels of certainty**, illustrated identically in Fig. 12 only through the mediator line, not through the level of physical confidence:

1. **Method 1** is established physics.
2. **Method 2** is a rigorous formalism, actively used in frontier experimental physics (LHC/MoEDAL monopole searches), but for an unconfirmed particle.
3. **Method 3** does not introduce a second fundamental charge (unlike Method 2) — the mediator M is a resonant mode of the field sourced by the **same** electric charge, not an independent magnetic charge coupled to its own B_μ. This makes it structurally simpler than Method 2, but also more speculative: its justification does not come from Dirac quantization as an a priori topological necessity, but from the TauTimp field's own, postulated spectrum.

The coincidence $\alpha_g \approx 34.25$ (Methods 2 and 3) is not accidental — it arises from **the same** Dirac quantization condition, applied identically to a minimally-charged monopole ($n = 1$). Independently, both methods solve non-perturbativity by restricting/modifying the effective channel: velocity-dependent coupling + magnetic-moment term κ (Method 2, s10052-018-6440-6, eq. 6–8, with the result that $\kappa = 1$ is the unique value restoring unitarity for spin 1) vs. the Bessel resonance rule + K-matrix unitarization (Method 3) — different mechanisms in detail, not formally equivalent, but convergent as a strategy.

What this document does **not** show: it does not derive the Bessel spectrum from QED or the Standard Model; Fig. 10–12 are conceptual representations of the Zwanziger formalism, not a re-derivation of the full 1971 Lagrangian; it does not compute observable cross-sections.

1.9 8. Conclusions

1. All three calculation methods for the p–e⁻ interaction discussed in this directory now have a complete Feynman representation (vertex + process diagram).
2. Method 2 (Zwanziger) receives, through Fig. 10–11, its first graphical representation in this document series — previously treated only through tables.
3. The bare Dirac coupling ($\alpha_g \approx 34.25$) is identical in Methods 2 and 3; each makes it tractable through its own, distinct mechanism.
4. Method 3 remains, structurally, distinct from a Zwanziger theory — a single fundamental charge, not two.
5. All rules in *Feynman.docx* remain satisfied.

1.10 9. References

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- D. Zwanziger, *Local-Lagrangian Quantum Field Theory of Electric and Magnetic Charges*, Phys. Rev. D 3 (1971) 880.
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- *monopole_wave_equation_full.md* — the Bessel spectrum $k_{r,n} = j_{0,n}/2R$.
- *Feynman.docx* — the original questions.
- *Feynman_ploturi_structurate_trei_metode.mpl* — companion Maple transcript, 3×3 dashboard for all three methods.

Maple app for QED:

https://michaelvio.github.io/Qed3metod.pdf